

Project Description

Name: Sports Car Price Analysis and Market Trends.

Problem Statement: The pricing of sports cars varies significantly across manufacturers, models, and performance specifications. Consumers, automotive analysts, and businesses often face challenges in understanding how different factors such as engine size, horsepower, torque, and acceleration influence the final price of a sports car. The main problem is to identify the key determinants of sports car pricing and analyze how performance-related variables impact price variations across different brands and models.

Possible Impact of Your Analysis: This research can provide significant information for a variety of stakeholders. Consumers can make more educated purchasing decisions by understanding which features warrant higher prices. It can help manufacturers and dealerships build pricing strategies and products by determining which performance aspects are most appreciated in the market. Furthermore, the analysis can help with data-driven decision-making in the automotive business by identifying trends, correlations, and patterns in sports car pricing. It may also help with predictive modeling, which allows you to estimate the price of an automobile based on its specifications.

Dataset(s):

The dataset used for this project is the Sports Car Prices Dataset, which contains information about various sports cars from different manufacturers.

The dataset includes the following variables:

Car Make: The brand or manufacturer of the sports car (e.g., Porsche, Lamborghini, Ferrari).

Car Model: The specific model or variant of the car (e.g., 911, Huracan, R8).

Year: The production year of the vehicle.

Engine Size (L): The engine capacity measured in liters, indicating performance potential.

Horsepower: The power output of the engine, influencing speed and acceleration.

Torque (lb-ft): The rotational force generated by the engine, affecting acceleration and handling.

0–60 MPH Time (seconds): The time taken for the car to accelerate from 0 to 60 mph, a key performance metric.

Price (USD): The market price of the car.

Source: <https://www.kaggle.com/datasets/rkiattisak/sports-car-prices-dataset>

To improve reliability, additional datasets will be explored to validate pricing trends observed in the primary dataset and to get more information.

Source: [https://www.kaggle.com/datasets/ayeshasiddiqua123/cars-pre/data?utm](https://www.kaggle.com/datasets/ayeshasiddiqua123/cars-pre/data?utm_source=baselight.app&utm_medium=referral&utm_campaign=kaggle-dataset)
[https://baselight.app/u/kaggle/dataset/metawave_vehicle_price_prediction?utm](https://baselight.app/u/kaggle/dataset/metawave_vehicle_price_prediction?utm_source=baselight.app&utm_medium=referral&utm_campaign=kaggle-dataset)